

Development and application of a molecular sexing protocol in the climate change-sensitive American pika

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Abstract Non-invasive sampling paired with molecular sexing provides a valuable tool for the study of elusive mammals, yet species-specific assays are required to ensure accurate identifications. The American pika (*Ochotona princeps*) is one example, having emerged as a focal mammalian species for studies of metapopulation dynamics and extinction risk in the face of climate change. Despite extensive study, knowledge of sex-specific patterns has been limited to small sample sizes given the need for live-trapping and field-based sexing. Here we describe a molecular sexing protocol designed to reliably determine sex from non-invasively collected hair. Our polymerase chain reaction-based method co-amplifies a male-specific sex-determining gene Y fragment and a species-specific autosomal microsatellite, the accuracy of which was validated on 15 internally-sexed individuals. Subsequent application to 157 hair samples demonstrated high rates of amplification success (96.0 %) and unambiguous sex determination (94.7 %), revealing a strong male-bias in the sample and one instance of inaccurate, field-based sex identification.

Keywords SRY · *Ochotona princeps* ·
Non-invasive · Conservation genetics

Traditional methods for sampling mammals include live-trapping, tagging, and morphological sexing prior to release (Wielgus and Bunnell 1994). Although effective for

many species, this method is not ideal for elusive mammals that are secretive by nature, typically resulting in low sample sizes (Banks et al. 2003; Piggott and Taylor 2003). Furthermore, some species lack external genitalia resulting in difficult live-sexing (Williams et al. 2004). Due to such barriers facing the study of elusive mammals, an alternative strategy is to combine non-invasive sampling and genetic approaches. This method facilitates increased sample sizes (Banks et al. 2003) and associated inferences of population history (Morin et al. 1993) and sex-specific patterns (Zgurski and Hik 2012), while minimizing impact on the sampled individual (Taberlet and Luikart 1999).

American pika (*Ochotona princeps*) are small, elusive mammals of conservation concern due to the threat of contemporary climate change (Smith et al. 2004). Applying non-invasive sampling and genetic approaches to the study of pikas is of great importance, as they are difficult to live trap, have poorly distinguished genitalia (Duke 1951), and are easily stressed (MacArthur and Wang 1974; Smith 1974). Previous work tested a polymerase chain reaction (PCR)-restriction fragment length polymorphism analysis of the sex-linked zinc finger gene (ZFX/ZFY) modified from Fontanesi et al. (2008), which had limited amplification success due to the large fragment size (~430 bp) and the degraded nature of the DNA that is characteristic of using hair as a starting material (Henry and Russello 2011). Here, we developed a molecular sexing protocol designed to reliably identify sex in American pika using non-invasively collected hair samples.

Our method uses a multiplex PCR to co-amplify fragments of the sex-determining gene Y (SRY; ~117 bp, male-specific) and a higher molecular weight (179–203 bp), species-specific autosomal microsatellite [Ocp10; Peacock et al., unpublished (Genbank Acc. No. GQ461705)], using a multi-tube approach (Taberlet et al. 1996). Molecular sexing

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using the SRY fragment has been previously employed in mammals (Palsbøll and Vader 1992; Vanpé et al. 2013); however, to our knowledge this method has not been applied to American pika. Our assay was designed based on the following parameters: (1) a small fragment size for SRY to enable amplification from degraded DNA; (2) a larger autosomal, species-specific fragment to act as a positive PCR control; and (3) sufficient size differential to enable agarose gel electrophoretic detection of multiple bands.

DNA extraction protocols for liver ($n = 7$) and hair ($n = 177$) samples followed Henry and Russello (2011) (see below for sample details). PCRs were carried out on an ABI Veriti thermal cycler in 12.5 μ L reactions containing: 5–10 ng DNA, 10 mM Tris–HCl, 50 mM KCl, 1.5 mM $MgCl_2$, 0.8 μ M of each SRY primer (Table 1), 0.8 μ M of each microsatellite primer (Table 1), 200 μ M dNTPs, 10 μ g bovine serum albumin, and 1U *AmpliTaq* Gold DNA polymerase (ABI). Cycling parameters were: 95 °C for 10 min, 40 cycles of 95 °C for 30 s, 48 °C for 30 s, and 72 °C for 30 s, and a final extension at 72 °C for 7 min. PCR products were run on a 1.5 % agarose gel containing 2.5 % SYBR Safe (Invitrogen). Samples with the *Ocp10* and SRY fragments present were males, and samples with only the *Ocp10* were females (Fig. 1).

This protocol was validated using samples from individuals of known sex and multiple starting materials (eight hair, seven liver), all of which were accurately diagnosed

using our method. The protocol was subsequently applied to 157 pika hair samples collected from a population in Oregon, USA that were previously identified as unique individuals of unknown sex through microsatellite genotyping (Robson 2013). The assay was repeated twice for each sample; in cases of incongruence, a third replicate was performed and a majority rule was applied. Overall, 150 (96.0 %) of the samples were successfully amplified, 142 (94.7 %) of which showed no inconsistency in sex determination between trials. Of the eight (5.3 %) samples that showed an inconsistency, four experienced allelic dropout of the SRY fragment between trials while the other four exhibited an incorrect size for the SRY lower molecular weight band. It should be noted that previously successful microsatellite genotyping of these samples might have reduced the occurrence of false alleles. The resulting sex ratio was male-skewed (60 % male: 40 % female); however, the degree to which this pattern is biologically meaningful or prone to ascertainment bias associated with the sampling method requires further study. This method was also applied to 12 hair samples collected from live-trapped individuals during an independent study of American pikas in British Columbia in order to confirm field-based sex identification. Of the 12 samples molecularly sexed, 11 were correctly identified in the field (data not shown). The lone discrepancy involved a juvenile, demonstrating the utility of the molecular test for

Table 1 Primers used in this study targeting *O. princeps*-specific Y-linked and autosomal regions

Region	Primer sequence	Size (bp)	References
Sex-determining region Y-chromosome (SRY)	F: AATGCATTCATACTATGGTC	117	This study
	R: CTCTGTAAGCTTTTCCACTG		This study
Autosomal microsatellite (<i>Ocp10</i>)	F: <u>TCCCAGTCACGACGTCCA</u> ATTGGCTGTTAAGGCTAT	179–203	Peacock and Kirchoff, unpublished (GQ461705)
	R: <u>GTTTCTTCCAGTGTCTGGC</u> ATACGGTAAGC		Peacock and Kirchoff, unpublished (GQ461705)

Underlined regions are non-template modifiers added to facilitate automated genotyping (Schuelke 2000; Brownstein et al. 1996)

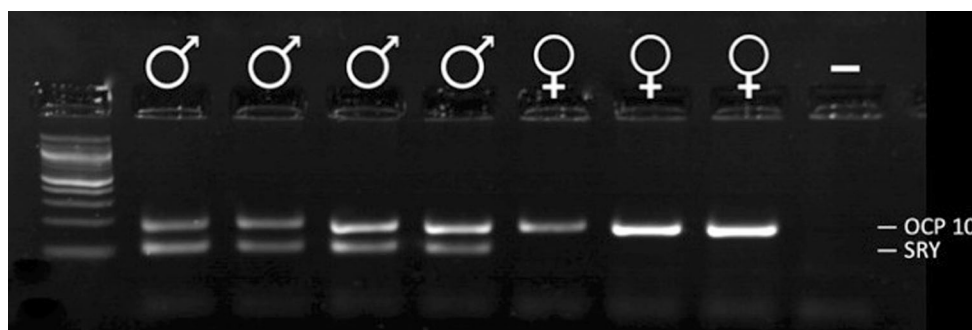


Fig. 1 Representative gel image used to identify sex in *O. princeps*, including males (♂), females (♀), 100 bp DNA ladder (NEB) and a negative control

confirming field-based sex identification (Blair, personal communication).

In summary, our method provided high rates of amplification success and reliable sex identification in the American pika from non-invasively collected hair samples. When paired with genotypic data, this molecular sexing approach will enable future studies of population and demographic history in this temperature-sensitive mammal of conservation significance.

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